

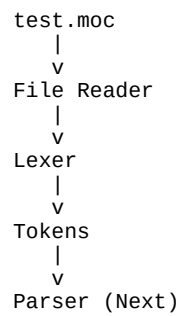
MyOwnCompiler - Day 04: Compiler Lexer V1

Compiler Engineering Journal

Day 4 Goal

Upgrade the lexer to read files, skip comments, track source positions, and produce clean token streams.

Compiler Pipeline



Features Added

- ✓ File Input
- ✓ Line Tracking
- ✓ Column Tracking
- ✓ Single-Line Comments
- ✓ Multi-Line Comments
- ✓ Comment Detection Logic

Bug Found & Fixed

Multi-line comments were being tokenized as SLASH and STAR tokens. The lexer was only checking comments once. The fix was to repeatedly consume whitespace and comments until real code was reached.

```
while(true)
{
    skipWhitespace();

    if(isCommentStart())
    {
        skipComment();
        continue;
    }

    break;
}
```

Verified Test Program

```
// variables

let x = 10;
let y = 20;

/*
multi line
comment
*/

print(x);
```

Verified Output

```
LET
IDENTIFIER
EQUAL
NUMBER
SEMICOLON

LET
IDENTIFIER
EQUAL
NUMBER
SEMICOLON

PRINT
LEFT_PAREN
IDENTIFIER
RIGHT_PAREN
SEMICOLON

EOF
```

Architecture Notes

Before:
source
current

After:
source
current
line
column

Interview Notes

1. What is lexical analysis?
2. Why remove comments in the lexer?
3. Why track line and column numbers?
4. Why is lexical analysis $O(n)$?
5. Difference between source text and token stream?

Revision Cheat Sheet

Day 1 - Project Setup
Day 2 - Basic Lexer
Day 3 - Operators & Symbols
Day 4 - File Input + Comments

Core Functions:
advance()
peek()
skipWhitespace()
skipComment()
identifier()
number()
tokenize()

Day 5 Roadmap

String Literals
Character Literals
Lexer Diagnostics
Unterminated Strings
Lexer Test Suite